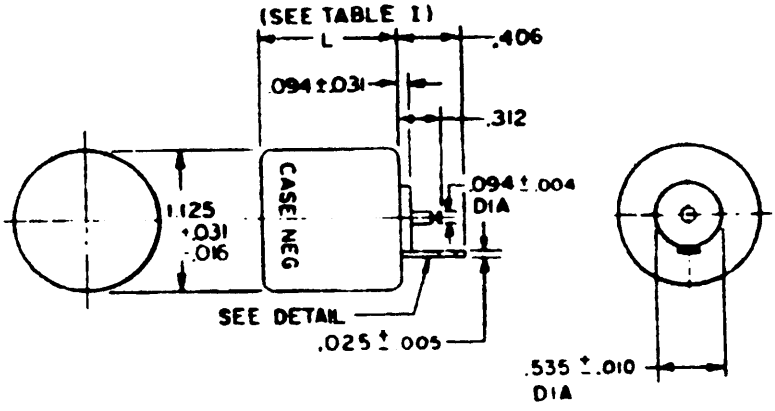
MILITARY SPECIFICATION SHEET

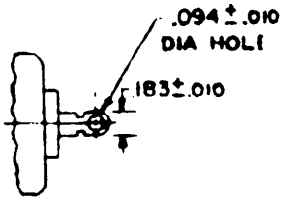
CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR17

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39006.

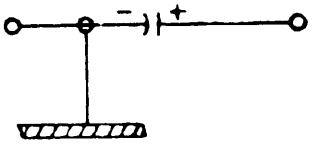
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.004	.10
.005	.13
.010	.25
.016	.41
.025	.64
.031	.79
.094	2.39
.183	4.65
.312	7.92
.406	10.31
.535	13.59
1.125	28.58



DETAIL OF SOLDER-LUG TERMINAL



CIRCUIT DIAGRAM

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.062 (1.57 mm).

FIGURE 1. Style CLR17 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)			Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Cap. change (-55°C)	Dimensions L x .063(1.59)	Part No. M39006/20- Failure rate level (S, J, 1,000 hr)		
			+25°C	+85°C	+125°C							L(2.0)	M(1.01)	P(0.1)	R(0.01)	S(0.001)
volts, dc	µF	%	µA	µA	µA	%	volts, dc	volts, dc	ohms	%	inches					
30	270	±20	10	135	180	10	39	34.5	15	-65	.500 (15.24)	1000	1050	1100	1150	1200
30	270	-50, -15	10	135	180	10	39	34.5	15	-65	.600 (15.24)	1001	1051	1101	1151	1201
30	650	±20	21	145	210	40	39	34.5	15	-85	1.100 (27.94)	1002	1052	1102	1152	1202
30	650	-80, -15	21	145	210	40	39	34.5	15	-85	1.100 (27.94)	1003	1053	1103	1153	1203
30	1000	±20	27	190	270	53	39	34.5	10	-85	1.100 (27.94)	1004	1054	1104	1154	1204
30	1000	-50, -15	27	190	270	53	39	34.5	10	-85	1.100 (27.94)	1005	1055	1105	1155	1205
60	200	±20	19	135	190	22	78	69.0	30	-50	.600 (15.24)	1006	1056	1106	1156	1206
60	200	-50, -15	19	135	190	22	78	69.0	30	-50	.600 (15.24)	1007	1057	1107	1157	1207
60	350	±20	23	155	220	37	78	69.0	25	-70	1.100 (27.94)	1008	1058	1108	1158	1208
60	350	-50, -15	23	155	220	37	78	69.0	25	-70	1.100 (27.94)	1009	1059	1109	1159	1209
60	700	±20	29	200	290	62	78	69.0	15	-70	1.100 (27.94)	1010	1060	1110	1160	1210
60	700	-50, -15	29	200	290	62	78	69.0	15	-70	1.100 (27.94)	1011	1061	1111	1161	1211
90	120	±20	19	135	190	13	78	103.0	40	-40	.600 (15.24)	1012	1062	1112	1162	1212
90	120	-50, -15	19	135	190	13	78	103.0	40	-40	.600 (15.24)	1013	1063	1113	1163	1213
90	220	±20	21	145	210	24	78	103.0	30	-60	1.100 (27.94)	1014	1064	1114	1164	1214
90	220	-50, -15	21	145	210	24	78	103.0	30	-60	1.100 (27.94)	1015	1065	1115	1165	1215
90	450	±20	29	195	280	45	78	103.0	35	-50	1.100 (27.94)	1016	1066	1116	1166	1216
90	450	-50, -15	29	195	280	45	78	103.0	35	-50	1.100 (27.94)	1017	1067	1117	1167	1217
180	82	±20	17	120	170	16	155	207.0	75	-40	.978 (24.79)	1018	1068	1118	1168	1218
180	82	-50, -15	17	120	170	16	155	207.0	75	-40	.978 (24.79)	1019	1069	1119	1169	1219
180	150	±20	19	135	190	15	155	207.0	60	-40	.978 (24.79)	1020	1070	1120	1170	1220
180	150	-50, -15	19	135	190	15	155	207.0	60	-40	.978 (24.79)	1021	1071	1121	1171	1221
180	300	±20	21	145	210	24	155	207.0	60	-60	1.938 (49.23)	1022	1072	1122	1172	1222
180	300	-50, -15	21	145	210	24	155	207.0	60	-60	1.938 (49.23)	1023	1073	1123	1173	1223
180	680	±20	29	200	290	46	155	207.0	50	-60	1.938 (49.23)	1024	1074	1124	1174	1224
180	680	-50, -15	29	200	290	46	155	207.0	50	-60	1.938 (49.23)	1025	1075	1125	1175	1225
270	82	±20	19	120	180	15	235	310.0	80	-40	1.350 (34.29)	1026	1076	1126	1176	1226
270	82	-50, -15	19	120	180	15	235	310.0	80	-40	1.350 (34.29)	1027	1077	1127	1177	1227
270	150	±20	19	135	190	22	235	310.0	100	-40	1.350 (34.29)	1028	1078	1128	1178	1228
270	150	-50, -15	19	135	190	22	235	310.0	100	-40	1.350 (34.29)	1029	1079	1129	1179	1229
270	270	±20	21	145	210	24	235	310.0	90	-60	2.812 (71.42)	1030	1080	1130	1180	1230
270	270	-50, -15	21	145	210	24	235	310.0	90	-60	2.812 (71.42)	1031	1081	1131	1181	1231
270	470	±20	29	200	290	45	235	310.0	75	-60	2.812 (71.42)	1032	1082	1132	1182	1232
270	470	-50, -15	29	200	290	45	235	310.0	75	-60	2.812 (71.42)	1033	1083	1133	1183	1233
360	120	±20	19	135	190	16	310	414.0	100	-40	1.705 (43.31)	1034	1084	1134	1184	1234
360	120	-50, -15	19	135	190	16	310	414.0	100	-40	1.705 (43.31)	1035	1085	1135	1185	1235
360	220	±20	21	145	210	22	310	414.0	120	-40	1.705 (43.31)	1036	1086	1136	1186	1236
360	220	-50, -15	21	145	210	22	310	414.0	120	-40	1.705 (43.31)	1037	1087	1137	1187	1237
360	390	±20	29	195	280	45	310	414.0	120	-40	1.705 (43.31)	1038	1088	1138	1188	1238
360	390	-50, -15	29	195	280	45	310	414.0	120	-40	1.705 (43.31)	1039	1089	1139	1189	1239
450	150	±20	19	135	190	16	390	518.0	150	-40	2.060 (52.83)	1040	1090	1140	1190	1240
450	150	-50, -15	19	135	190	16	390	518.0	150	-40	2.060 (52.83)	1041	1091	1141	1191	1241
450	270	±20	19	135	190	23	390	518.0	150	-40	2.060 (52.83)	1042	1092	1142	1192	1242
450	270	-50, -15	19	135	190	23	390	518.0	150	-40	2.060 (52.83)	1043	1093	1143	1193	1243
540	82	±20	17	120	170	16	470	621.0	180	-40	2.435 (61.85)	1044	1094	1144	1194	1244
540	82	-50, -15	17	120	170	16	470	621.0	180	-40	2.435 (61.85)	1045	1095	1145	1195	1245
540	150	±20	19	135	190	22	470	621.0	170	-40	2.435 (61.85)	1046	1096	1146	1196	1246
540	150	-50, -15	19	135	190	22	470	621.0	170	-40	2.435 (61.85)	1047	1097	1147	1197	1247
830	82	±20	17	120	170	16	545	724.0	190	-40	2.810 (71.37)	1048	1098	1148	1198	1248
830	82	-50, -15	17	120	170	16	545	724.0	180	-40	2.810 (71.37)	1049	1099	1149	1199	1249
830	150	±20	19	135	190	23	545	724.0	200	-40	2.810 (71.37)	1050				
830	150	-50, -15	19	135	190	23	545	724.0	200	-40	2.810 (71.37)	1051				

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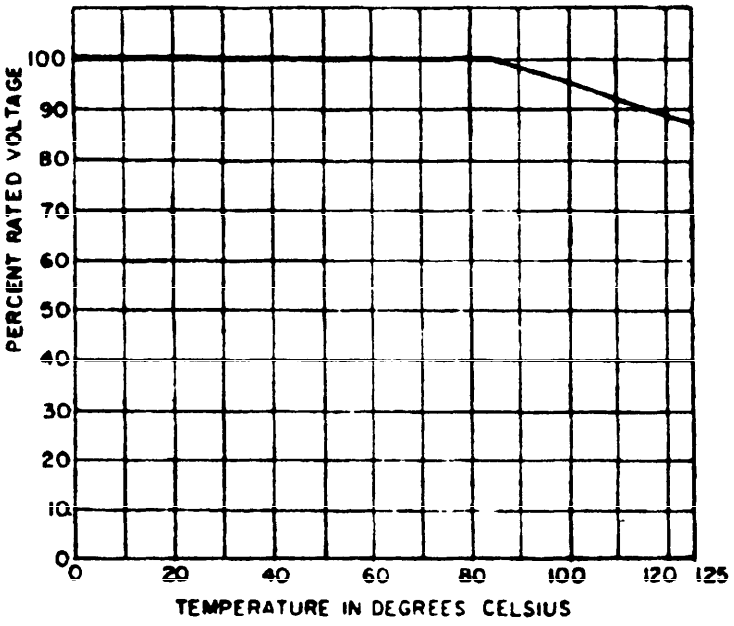


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, uninsulated.
Seal type - G (hermetic).
Terminals - Solder-lug.
Rated temperature - -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - See table I.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Not applicable.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Bend (if required) - Method 211 of MIL-STD-202, condition B.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

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Dielectric withstanding voltage (DWV):
Not applicable.

Insulation resistance (IR): Not applicable.

Low temperature (storage): Method 502 of MIL-STD-810.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL - See table I.
Cap. - Within tolerance of table I value.
DF - See table I.
Step 2 (-55°C):
Imp. - See table I.
ΔCap. - Within percent shown in table I of step 1 value.
Step 3 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.
Step 4 (+85°C):
DCL - See table I.
ΔCap. - +20% of step 1 value (max).
DF - See table I.
Step 5 (+125°C):
DCL - See table I.
ΔCap. - +25% of step 1 value (max).
DF - See table I.
Step 6 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.
2,000-hr (qualification):
DCL - See table I.
ΔCap. - Within ±20% of initial measurement.
DF - See table I.
240-hr (group B):
DCL - See table I.
ΔCap. - Within ±10% of initial measurement.
DF - See table I.
10,000-hr (group C) and extended life:
DCL (85°C) - 125% (max) of table I value.
DCL (25°C) - See table I.
ΔCap. - Within ±25% of initial measurement.
DF - 200% (max) of table I value.
DWV and IR - Not applicable.

Vacuum life: Not applicable.

Life at 125°C: Method 108 of MIL-STD-202.
DCL - See table I.
ΔCap. - Within ±20% of initial measurement.
DF - 150% (max) of table I value.
DWV and IR - Not applicable.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
Test voltage - Rated voltage for 30 to 450v units; 500 vdc for 540 and 630v units.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Marking: Capacitors shall be marked as shown in the following example:
EXAMPLE:

M39006/20-1000	- Part number.
JAN "JM"	- "JAN" brand and trademark.
12345 7015A	- Source code, date code, and lot symbol.
370μF 30VDC	- Capacitance and rated voltage.

Polarity - Capacitors shall be marked "CASE NEG" on the negative end (see figure 1).

Voltage groups (85°C rated voltage):

I	30 to 90v incl
II	180 to 360v incl
III	450 to 630v incl

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Custodians:
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Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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